

Supplementary file

1 Introduction

The proposed model BioEGAT is accompanied by an interactive Gradio web application designed to provide researchers with an intuitive, transparent interface for biomedical link prediction. The web application allows users to explore the pipeline’s predictions, inspect retrieved structural evidence, and analyze the reasoning trace of the Large Language Model (LLM). This document provides a visual walkthrough of the interface and its core functionalities.

2 Module 5: Explanatory Context and LLM API Call

Module 5 functions as an explanatory layer. Rather than acting purely as an isolated reasoning engine, this module parses the structural evidence retrieved by the KGE and provides critical supplementary information and context for both the human user and the LLM API call.

2.1 Path Unrolling and Prompt Construction

To bridge the gap between mathematical graph embeddings and the natural language capabilities of the LLM, Module 5 performs evidence unrolling. It translates the multi-hop mechanism paths retrieved from the KGE into human-readable text sequences (e.g., Disease --[associated with]--> Gene --[target]--> Drug).

These unrolled paths are then fed directly to the demo page.

Candidates (BioEGAT rerank order):

1. {candidate_1}
 - Unrolled Paths:
 {query} --[{rel}]--> {intermediate_node} --[{rel}]--> {candidate_1}
 - ...
2. {candidate_2}
- ...

By injecting these parsed evidence paths, the demo is shown to synthesize the KGE’s structural connectivity with additional medical knowledge.

2.2 Statistical Grounding: PSR and Hypergeometric p-value

In addition to unrolling paths, Module 5 computes statistical metrics to quantify the reliability of the retrieved evidence. These metrics, the Path Support Rate (PSR) and the hypergeometric p-value, are displayed to the user to provide transparent context for the LLM’s final decision.

Path Support Rate (PSR) The PSR score quantifies the relative density of structural evidence supporting a candidate. It is computed as the number of valid, multi-hop mechanistic paths (up to $k = 3$ hops) connecting the query entity to the specific candidate, divided by the total number of paths extracted in the query’s semantic neighborhood subgraph.

- **Range:** The PSR score ranges from 0.0 to 1.0.
- **Interpretation:** A good candidate typically exhibits a PSR score between 0.1 and 0.2, depending on the overall density of the local subgraph. However, the absolute value is less important than the relative ranking; candidates with the highest PSR scores within the top-20 list possess the strongest structural backing from the knowledge graph.

Hypergeometric p-value To ensure the retrieved paths are statistically significant and not merely artifacts of highly connected hub nodes, the pipeline computes a hypergeometric p-value. This metric compares the observed number of supporting paths against the expected number of paths if edges were distributed randomly across the candidates.

3 Configuration and Query Setup

The image shows a web-based configuration interface titled "1. Configuration". It contains two main sections: "Dataset" and "LLM Settings".

Dataset Section:

- Dataset:** A dropdown menu with "PrimeKG" selected.
- Query entity:** A dropdown menu with "Atenolol" selected. Below it is the text "The head entity to complete from".
- Prediction direction:** A dropdown menu with "Predict tail (head + relation → ?)" selected. Below it is the text "Predict the head or the tail of the triple".
- Relation:** A dropdown menu with "indication" selected.

LLM Settings Section:

- Enable Live Gemini API Call:** A checkbox that is checked. Below it is the text "Uncheck to display pre-computed LLM predictions instantly".
- Gemini Model:** A dropdown menu with "gemini-2.5-flash" selected.

Figure 1: Configuration interface for selecting the dataset and query entities.

As shown in Figure 1, the left panel of the demonstrator serves as the primary configuration zone. Users can select from multiple supported datasets (e.g., PrimeKG, Hetionet). Following dataset selection, the user specifies a query entity (e.g., *disease*), the prediction direction, and the relation type (e.g., *indication*). The interface also provides options to toggle live Gemini API calls, allowing users to either query the LLM in real-time or fall back to pre-computed predictions for faster exploration.

4 Evidence and Ranked Candidates

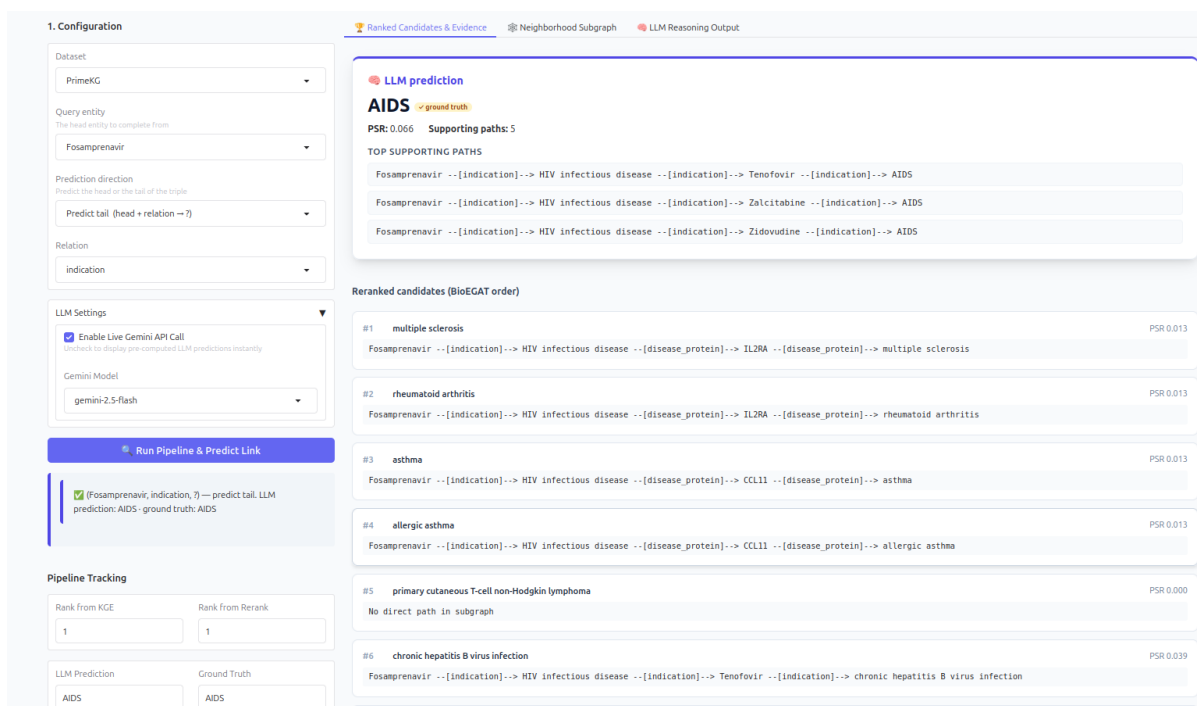


Figure 2: The Ranked Candidates tab displays the top predictions and their associated Path Support Rate (PSR) scores.

Figure 2 illustrates the primary results interface. Upon completion of the processing workflow, the system presents a prominent information card highlighting the top-predicted entity (the rank-1 candidate). Below this, a comprehensive table lists the top 20 candidates, sorted by their LLM-assigned likelihood. Each candidate is annotated with its Graph Neural Network (GNN) rerank certainty and its Path Support Rate (PSR) score, which quantifies the volume of mechanistic evidence retrieved from the knowledge graph. This transparent ranking allows domain experts to quickly assess the statistical confidence behind each prediction.

5 Pipeline Execution and Tracking

The interface is titled "Pipeline Tracking" and contains four input fields arranged in a 2x2 grid. The top row shows "Rank from KGE" and "Rank from Rerank", both with the value "1". The bottom row shows "LLM Prediction" and "Ground Truth", both with the value "glioblastoma (disease)".

Figure 3: Pipeline tracking interface showing the intermediate ranking steps.

Figure 3 shows the intermediate tracking steps of the pipeline, where users can see the initial rank from the KGE, the reranked position from the GNN, and the final decision from the LLM compared against the ground truth.

6 Neighborhood Subgraph Visualization

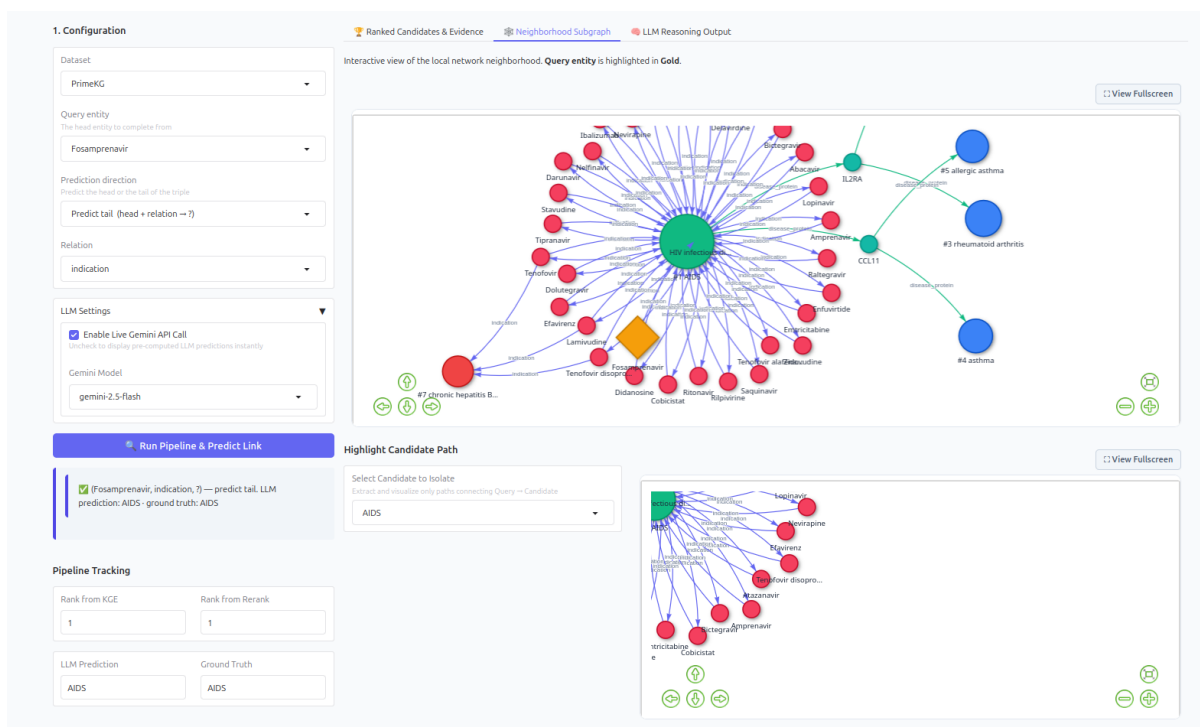


Figure 4: Interactive PyVis visualization of the full semantic neighborhood subgraph.

A critical feature of the BioEGAT demonstrator is its ability to visualize the structural evidence used to make predictions. As seen in Figure 4, the “Neighborhood Subgraph”

tab renders an interactive, physics-based graph using the PyVis library. The node styling provides immediate semantic context:

- **Query Entity:** Highlighted in **Gold** to serve as the visual anchor.
- **Proteins/Genes:** Colored in **Teal**.
- **Diseases/Phenotypes:** Colored in **Rose**.
- **Drugs/Compounds:** Colored in **Violet**.
- **Candidates:** Shaded based on their GNN certainty score (darker indicates higher confidence).

Users can pan, zoom, and drag these color-coded nodes to explore the multi-hop mechanistic pathways bridging the query disease to potential drug targets.

7 Candidate Path Isolation

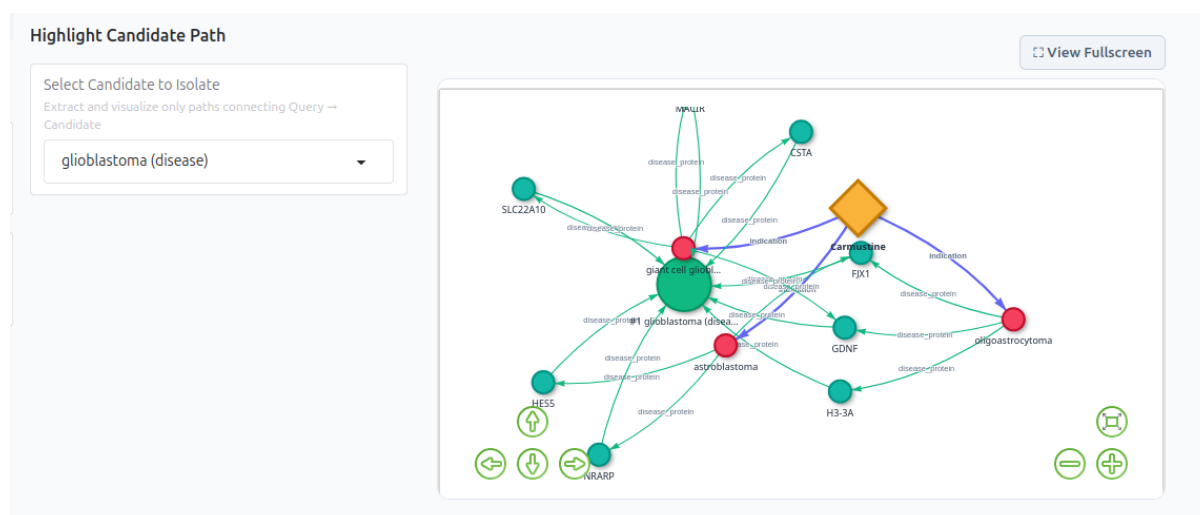


Figure 5: The highlighted path: choosing a candidate isolates the mechanistic paths from the query to that specific candidate.

To reduce visual clutter in dense semantic neighborhoods, the interface provides a candidate path isolation tool. As shown in Figure 5, when a user chooses a candidate from the dropdown, the system highlights and displays *only* the shortest paths connecting the query entity to that candidate. This feature is invaluable for targeted evidence review, allowing researchers to isolate and interpret the specific genetic or phenotypic bridges supporting a given drug-repurposing hypothesis.

8 API Call’s Results: Out-of-Graph Generalization

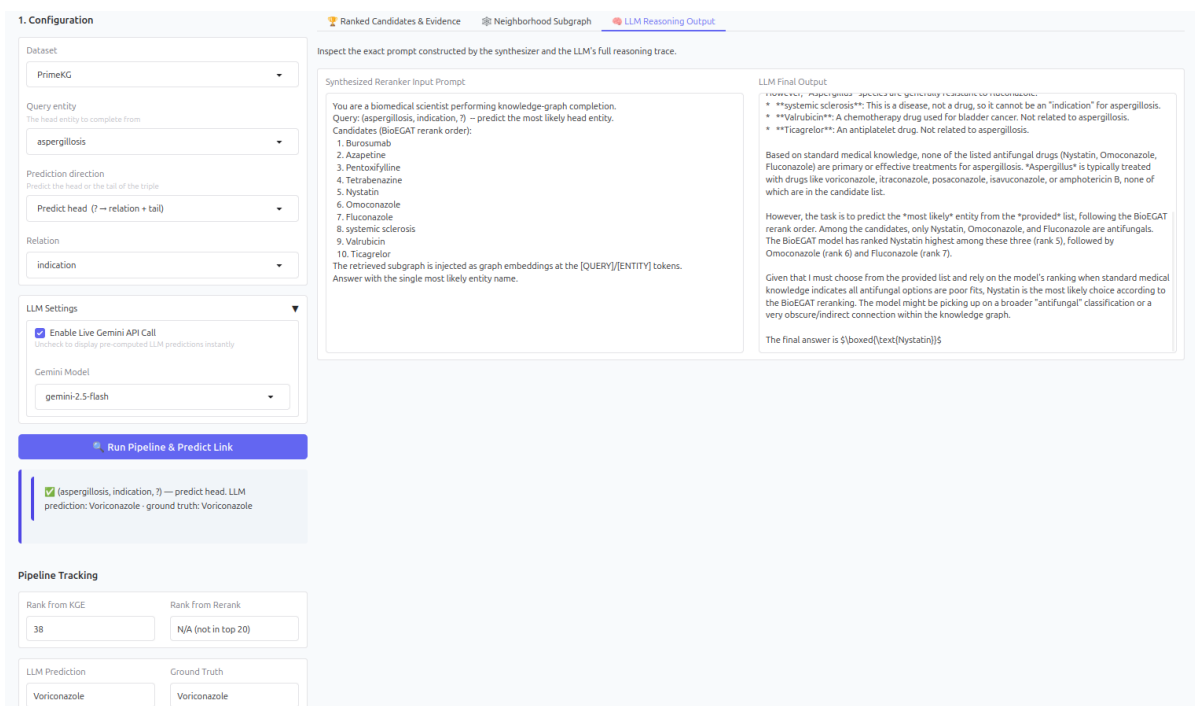


Figure 6: The API Call’s Results tab: the optional live Gemini hypothesis channel surfaces Voriconazole, a target the KGE retriever missed.

Figure 6 displays the “API Call’s Results” tab, which provides complete transparency into the LLM’s decision-making process. The left pane shows the exact prompt constructed by the pipeline, while the right pane displays the LLM’s raw reasoning trace.

Crucially, this image highlights an optional capability of the demonstrator: out-of-graph candidate generation via the live Gemini API call. In this specific query, the KGE (Knowledge Graph Embedding) module failed to retrieve the ground-truth target, *Voriconazole*, leaving it out of the coarse candidate list and the structural evidence paths. Because the scored reranking pipeline operates only on the fixed KGE top-20 candidates, it cannot by itself recover a target that was never retrieved.

The web application therefore offers an optional zero-shot hypothesis channel: when live Gemini calls are enabled, an external LLM proposes additional candidates from its parametric medical knowledge, which are then fuzzy-linked back to graph entities and fused with the KGE list. Here that channel surfaced Voriconazole as the correct entity.